

Exploring Tangible User Interface Design for Social Connection Among Older Adults: A Preliminary Review

Pallabi Bhowmick
pbhowmic@iu.edu

Indiana University Bloomington
Indiana, USA

Erik Stolterman Bergqvist
estolter@indiana.edu

Indiana University Bloomington
Indiana, USA

ABSTRACT

Older adults (aged 65 or above) are the fastest growing section of the population and increasing number of older adults are reporting of social isolation. WHO recognizes social engagement as an important factor for reducing isolation among older adults. Researchers have historically investigated digital technologies to build connections, but they involve disadvantages like accessibility issues and have not been a good match for the aged population. In this paper, we investigate existing work that focuses on improving social connections among older adults and then explore the potential of tangible user interaction as an alternate solution. We conclude the paper with design consideration for HCI researchers working in the space of tangible user interface design for older adults. This paper is a preliminary review towards opening a conversation about the suitability of tangible design for older adults in improving social connection.

CCS CONCEPTS

• Human-centered computing → User interface toolkits.

KEYWORDS

older adults, social isolation, tangible technologies

ACM Reference Format:

Pallabi Bhowmick and Erik Stolterman Bergqvist. 2023. Exploring Tangible User Interface Design for Social Connection Among Older Adults: A Preliminary Review. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3544549.3585722>

1 INTRODUCTION

Older adults are now the fastest-growing population in the United States [18], they have far-reaching social impacts, and this has led to a growing attention towards aging-related concerns and providing support to the older adults from the HCI research community [46, 76, 90, 105]. Social isolation among older adults have become a growing concern owing to a shrinking social circle stemming from retirement, illness, or bereavement [67]. Institutionalization becomes imperative with declining mental, physical health, and increasing safety issues with age [48]. Older adults want to

avoid institutionalization and desire to age in their own home [50]. HCI researchers have explored various technologies to help older adults age-in-place such as, sensor-equipped smart homes [102], location tracking apps [129], physical in-home systems [128] or wearables [108] to monitor health parameters and share well-being information to care providers or loved ones. However, such passive monitoring takes away agency and control from older adults, which is dehumanizing and gives rise to privacy issues [34, 97].

While a large corpus of works have focused on the medical health of older adults, improving social connection plays an important role in reducing isolation [95]. The Harvard Longitudinal Study of Adult Development [100], which is almost 80 years old and one of the world's longest studies, shows that embracing community helps us live longer, and happier. One key to healthy aging is close, meaningful relationships [144]. With the advent of digital technologies, communication has become easier, but digital technologies do not offer an intuitive, accessible medium for older adults, resulting in a steeper learning curve [10, 79]. Furthermore, the biometrics or tactile feedback do not work as people age [55], because the firmness and elasticity of skin reduces and fine wrinkles appear in the skin [124]. Tangible systems, however, could provide more accessible systems leveraging the use of existing competences and practices through common, familiar objects [79, 123, 136]. Tangible systems also meet the aesthetic needs of older adults resulting in better adoption of technology [91, 106].

In this paper, we investigate how previous research have addressed topics on older adults and social isolation, how researchers have tried to improve social connections among older adults, and then explore the possibilities and potential of tangible technologies in this area of research. The aim is to explore these topics from a technological standpoint. While digital technologies have connected the world, they have not been a good fit for the older adult population because they are not developed keeping the needs of older adults in mind. This paper, through a preliminary review of tangible technologies, will explore their use and advantages over digital technologies for the aging demographic. Finally, this paper will suggest design considerations for HCI researchers designing tangible user interfaces for older adults.

2 METHODOLOGY

Our work was motivated by the following research questions - a) What technologies are researchers focusing on when building for older adults to address isolation? and b) What are the challenges with these existing technologies and how can tangible technologies potentially emerge as an alternate solution? This preliminary review is driven by the motivation to explore the current state-of-the-art in tangible user interface design for older adults and identify

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585722>

the potential of tangible design for this demography. The ACM Digital Library was searched by the first author with the keywords, ("*older adult*" OR "*elder*" OR "*senior*" OR "*aged*") AND ("*isolation*" OR "*loneliness*") since the year 2000, which returned 3502 results. Papers were screened based on title, abstract, and full text. All papers that did not describe any technology use, design, or development, were excluded. The decision to not include "tangible" as a search keyword, enabled us to explore a broader scope of research on technologies that are designed to alleviate social isolation among older adults. Papers were then filtered based on whether the technologies were tangible or not. Any technologies that were screen-based and did not include a tangible physical component that older adults could interact with, were excluded. We found 115 papers that explored the use of tangible user interfaces for supporting aging-in-place and reducing loneliness among older adults (e.g., using robots as companion pets [26]), out of which only 23 papers focused on using tangible user interfaces primarily to promote social connection for older adults with others. In this preliminary review, papers on tangible technologies were investigated to investigate what made tangible user interfaces particularly suitable for the older adult population over existing technologies.

3 RELATED WORK

Population aging is a global phenomenon. In US alone, it is expected that by 2050, the older adult population will be 21.4% of the total population [88]. Aging is a long process that spans across an individual's life from birth to death. While biological processes play an important role, it is also an individual's social and environmental conditions that contribute to aging [41]. According to a report from the National Academies of Sciences, Engineering, and Medicine (NASEM), 24% older adults in USA are considered to be socially isolated [109]. Older adults often report concerns about social isolation [148], which arises from a feeling of lack of social connectedness [92]. Social isolation is defined as a lack in quantity and quality of social contacts [12] resulting from psychological or physical distancing, or both, from an individual's desired network of people.

In the late 1990s, the World Health Organization adopted the term "active aging" to convey a more inclusive message about aging [115]. Active aging is an umbrella concept that focuses on improving quality of life for older adults, from both physical and social perspectives [13]. According to WHO, "Ageing takes place within the context of others – friends, work associates, neighbours and family members." [115]. Therefore, a two-way dependence is an important tenet for active aging [47, 127]. Maintaining meaningful social relationships and contacts are key components for quality of life [13, 19, 51, 144]. A shift in social dynamics can be observed in the twenty-first century with older adults living farther away from their family [122]. Several other reasons that contribute to older adults' disconnection from society include, gradual development of age-related mobility issues, or mobility restrictions due to injury/disability [150], decline in health causing poor vision or loss of hearing [142], occurrence of diseases that can cause stigma [12], and traumatic events such as death of spouse [33]. Consequences of social isolation include poorer mental health [71, 87, 89, 137] and an increased risk of mortality [30, 36, 137].

As people age, they tend to lose agency and others (families, professional caregivers) start making decisions for them. This becomes especially prevalent when older adults are no longer able to perform daily tasks such as ADLs and IADLs independently (Activities of Daily Living (ADL) [4], that are fundamental to basic survival such as, eating, bathing, toileting, dressing, etc. and Instrumental Activities of Daily Living (IADL) [35], that are more complex activities like shopping, meal prep, house work, managing finances, transportation, communication, medication, etc.) [53, 115]. When older adults experience limitations in maintaining their home, they are more likely to be institutionalized in assisted living and care facilities [7]. Moving to a new facility can also restrict an older adults' autonomy, increase cost of care, and reduce quality of life due to negative psychological effects of institutionalization [63, 82]. According to the latest American Association of Retired Persons (AARP) survey, older adults express aging in place as a primary goal [148].

Most older adults prefer to age independently in the familiar setting of their own homes [50, 62, 134], and smart homes equipped with environmental, in-home, and wearable sensors can help achieve the same [2, 21, 49, 64, 84, 153], by monitoring daily behavior and activity patterns of older adult residents. While some studies indicate older adults typically report an enhanced feeling of safety [117, 133], there is a growing concern that these technologies will actually replace human contact, in-person interactions, and give rise to isolation among older adults [97, 99]. Such passive monitoring systems also take away agency, and research [97, 130] suggest that older adults prefer to be active users of the system instead of being passively monitored. They prefer systems that can provide data transparency, agency and control, and are non-intrusive in nature.

Most research on smart homes typically focus on the physiological aspects of aging since physical safety is one of the biggest challenges older adults face [117]. However, Liu et al. [95] says this idea needs to change because, over the years with improved medical and living standards, older adults remain healthier for a longer period of time. The need for better social and emotional well-being is in fact getting stronger, which in turn could have a positive effect on physical health.

3.1 Digital Technologies

HCI researchers had started using social technologies a couple of decades back to alleviate concerns of isolation among the aging population [68, 106]. Chui et al. [25] defines social technologies as "digital technologies used by people to interact socially and together to create, enhance, and exchange content". Increasing use of social technology is associated with better subjective health and well-being in older adults [23] and Internet usage can potentially result in mental health benefits [31, 32], along with a sense of social inclusion [139]. In the recent decade, use of Information and Communication Technologies (ICTs) have become very prevalent for information gathering and social connection [151]. Social networking sites, video-conferencing apps have made long-distance communication easy and helped people stay connected [9, 83, 93, 131, 140]. Tele-care facilities such as Care TV [145] have been observed to improve social connectedness and well-being [10, 32].

Researchers have also developed digital systems to tackle issues of social isolation. Neves et al. [10] developed an iPad-based app to support asynchronous communication of older adults with their family and friends. They found increased interaction and perceived sense of connection among the participants. Garattini et al. [58] observed similar positive responses using the Building Bridges system, a touch-screen computer system to foster communication and reduce isolation. Video games, such as playing Wii, have also been found to stimulate positive feeling, social interaction, and reduced isolation [81]. Czaja et al. [39] developed a software application, PRISMS for older adults to support a wide array of features including social connectivity. Participants using PRISMS reported higher engagement, better social support, and reduced loneliness [38].

While the rise in ICTs and digital technologies have made lives easier, their adoption has been slow among the aging population. Common reasons include inaccessibility, usability issues, security concerns, or high costs [24, 65, 79, 136]. Existing literature emphasizes the need for making digital communication systems more accessible for this age group [5, 58] to bridge "digital divide" [20, 114]. Spreicer suggests that biological aging leads to reduced physical and functional capabilities, and cognitive aging reduces information processing abilities making it more difficult to learn and adapt new technologies, such as screen-based interactions [136]. Neves et al. [113] developed InTouch to be an accessible and tailored communication app, yet there was a learning curve and additional training sessions were needed to bridge the gap. Older adults avoid technologies they believe are too difficult to learn [86]. Owing to barriers stemming from limited experience of using new technologies, older adults often find digital interfaces non-intuitive and complex [40, 136].

Need for web proficiency, knowledge about social media and online social norms, put older adults at a disadvantage [10, 37, 45, 75, 152]. A longitudinal study [14] states loneliness among older adults increased from using social networking sites, because of anxiety from poor engagement. The author suggests the idea that only "socializers" benefit from social media. Non-intuitive, unfamiliar interfaces leading to a higher learning curve act as a major barriers to technology adoption [38, 58, 112]. Furthermore, the needs of the "oldest old" or frail older adults are particularly not considered in the design, deployment, and study of such systems [112]. Findings suggest that proper training to use mobile technology can increase older adults' technological literacy, and in turn, technology adoption [45, 75, 152].

Recognizing the necessity of accounting for biological, cognitive, and social aging in technology design is imperative. Spreicer et al. [136] showcases two instances of tangible communication technologies to illustrate the potential of tangible interfaces as a workable option for senior citizens. One is the marble answering machine that triggers a message when a user drops a marble at a specific spot, while the other is TanCu, which enables messaging by placing the appropriate side of a cube on a designated location. Tangible technologies provide certain advantages over digital screen-based interfaces [79, 135]. Tangible interfaces incorporate physical objects as a way to interact with digital information or systems, thereby offering familiarity and intuitiveness to users. By allowing tactile manipulation, it has the potential to bridge the gap between physical and digital worlds, thus minimizing digital

divide. It can eliminate the necessity for screen-based interfaces that require cognitive exertion or dexterity, thereby catering to the biological and cognitive aspects of aging. Tangible technologies therefore have the potential to increase technology adoption, and when used as social technologies, can help reduce isolation among older adults. In this preliminary review, we will investigate the potential of tangible technologies as an alternate solution to improve social connection for older adults.

4 FINDINGS

In this section, we will discuss the findings from the preliminary review investigating the affordances of tangible user interfaces that make them particularly beneficial for older adults and the potential of tangible technologies to emerge as an alternate solution for the older adult population.

4.1 Tangible Technologies

Aging-in-place technologies have been developed to help older adults maintain their independence longer, but their acceptance is limited [118]. More often, needs or values of older adults are not taken into consideration while building a technology which eventually results in non-adoption because older adults do not find it relevant or useful [27, 52, 101]. Davis's Technology Acceptance Model (TAM) [42] states two important factors behind technology acceptance by user: perceived ease of use and perceived usefulness. The former refers to "the degree to which an individual believes that using a particular system would be free of physical and mental effort", while the latter is defined as, "the degree to which an individual believes that using a particular system would enhance his or her job performance". This model was adapted and updated by several researchers and later Kim et al. [85] created a TAM specifically for older adults. Researchers further suggests that older adults avoid technology that they perceive as being too difficult to learn, not meaningful, scalable, or customizable [65, 116].

The influential paper on "Tangible Bits" by Ishii et al. [77] opened new opportunities of interaction with tangible, physical objects. Their paper first proposed a move beyond the dominant Graphical User Interface (GUI) to Tangible User Interface (TUI) by augmenting real physical objects with digital information. There has been some relevant research so far to address the challenges of traditional screen-based options with tangible solutions for older adults. Joshi et al. [79] developed GLiMT, a tangible alternative in the form of a wooden cube to assist older adults who struggled with traditional screen-based solutions in a care home. GLiMT opened up new and promising opportunities for interaction because it leveraged familiar inputs and movements resulting in a less complex, more intuitive option for the elderly. Researchers believe "non-tech" appearance of a system can improve acceptability among the aging population [147]. Varesano et al. introduced PALLA, a wireless spherical input device equipped with sensors, but without any buttons or switches so that it looks and can be used like any other non-technological everyday object [147]. PALLA was well-accepted and participants found PALLA to be amusing and engaging.

A common research topic for the aging demographic involves medication management, which is one of the chief ADLs for enabling aging-in-place. Researchers have used tangible components

in improving medication adherence to ensure it is comprehensible to older users [8]. Examples of such systems include aesthetically-pleasing ambient systems [59, 91, 126], digitally or sensor-augmented smart pillboxes [44, 91, 126, 138], RFID-enabled medication monitoring system integrating wireless sensors and a GUI [70]. These research have showed that such systems are beneficial for older adults to maintain their medication intake, and in turn, their health.

Another common topic of research involves assisting older adults with cognitive impairment. Gamberini et al. [56] developed Eldegames, an interactive tabletop gaming platform that leverages use of familiar objects such as pens as input devices to assist older adults in cognitive training and preserving cognitive functions. There are other similar examples of using tabletop game [80] and tangible interface integrated with NFC tag in a game setting [43] to train cognitive abilities. Some researchers have used exergames with tangible components such as, Nintendo Wii, SilverBalance [60] and SilverPromenade [61]. It comes with a handheld controller equipped with sensor and can be used as assistive technologies in rehabilitation to improve balance, gait, and muscle strength, or even to reduce depression [143].

Researchers have used tangible technologies to support other age-related challenges and prolong independent living. Murata et al. [104] developed a wearable "self-gait training assistant system" to monitor gait of the older adult user and provide assistive information using a mobile floor projector while walking. Yang et al. [154] developed SITUG, a wearable sensing device that can predict fall risk through advanced gait monitoring. Researchers have also used tangible assistive technologies for assisting ADLs such as bathing [98], or assistive robots to perform home upkeep tasks such as cleaning and organizing [11], or a robotic medication reminder assistant [141], or even as emergency response [57]. Assistive systems such as, DALi intelligent walker [17], Smart Walker [22], Smart Wheelchair [149], intelligent rehabilitation chair [155], assistive wheelchair robots [132], are becoming increasingly common to assist with mobility and improve stability. Each of these systems leverage familiarity by incorporating functionalities to objects that are already habituated to.

4.2 Tangible Technologies for Social Connection

Since early 2000s, researchers have been creating tangible, ubiquitous technologies for older adults [28, 69, 106]. Early work that sparked interest in tangible technologies for social connectedness include papers by Mynatt et al. [106] on Digital Family Portrait and Hindus et al. [68] on social communication devices at home. Researchers have investigated the use of electronic systems to help older adults to tackle the issue of isolation by sharing well-being information. These physical electronic systems are user-facing systems that require participation through one-way or two-way communication. Digital Family Portrait [106] could connect older adults with their family members through one-way communication. It consisted of an ambient display that conveyed regular activity data of older adults from their daily lives through sensor data. This data is then sent to the loved ones of older adults, thereby giving them the necessary mental peace. Concerned family members could gain remote access to daily well-being information of the older adult just

by glancing at the display. CareNet [28], another one-way communication system, offered an ambient display to adult children of an older adult to monitor various activities and events (outings, falls, medication, etc.) of the older adult. The ABRA system [119] allowed family to receive well-being data of their senior member through physiological and motion sensor while also supporting communication. BUMP, a video-based networking tool by Chu et al. [24] used PIR sensors for passive motion detection and information transfer between older adults and family.

However, one-way communication tends to take away agency from older adults by including them only as passive recipients of care, and not as active users. As part of the Casablanca project, Hindus et al. [68] experimented several prototypes and design concepts, all of which leveraged the need for older adults to connect with friends and family through two-way communication, by either providing shared tangible spaces for asynchronous communication (CommuteBoard, MessageBoard, ScanBoard), or ambient awareness prototypes (InTouch, Presence Light, Intentional Presence Lamp [IPL] - Curtain IPL, Lampshade IPL) that could indicate a user's presence. Plaisant et al. [121] developed a shared digital paper calendar offering two-way information sharing, wherein older adults write with a digital pen, which transfers data when the pen is inserted in the cradle. InSight [96], a system for reciprocal communication based on the idea of interacting through a fogged up glass panel was accepted better than pixelated displays because of its familiarity and intuitiveness. Piper et al. [120] developed an interactive photo album to stimulate intergenerational social communication. Baharin et al. [6] deployed SonicAIR, an ambient awareness technology probe that could relay auditory information between participants in two locations via Twitter feed. Researchers also explored technology, such as the Presence Clock [74], that allowed two-way information sharing. This system connected older adults and their caregivers and required active participation from older adults, thereby enabling them to have more control and independence. The system involved a pair of analog clocks with LEDs that could light up on detecting activity and the intensity of the brightness indicated the amount of activity on either end. This way older adults could check-in on their family caregiver and vice-versa. The Check-in Tree [3] offered a peer-to-peer morning check-in functionality to connect older adults with their loved ones.

Other research works on encouraging social engagement include sensors-equipped Smart TV [66], Social Yoga Mats [107], MoviPill [44] containing an app-connected physical pillbox to create a social circle of older adults competing with each other over medication adherence, or games to support inter-generational play [146]. The design of Memento [94] takes inspiration from slot machines and facilitates inter-generational communication. Messaging Kettle, a smart communicating device by Soro et al. [135] and Nettle [54] serves as other good examples of such systems that leverage traditional routine of making tea to improve social connectedness among the aging population. It may be observed that most of these works leverage using familiar objects to successfully achieve research goal. This circles back to our initial argument about building on older adults' emotional attachment and familiarity to everyday objects or "habituated objects" [15] to increase technology adoption among this demographic. In 2006, Hornecker et al. [73] stated that "The support of social interaction and collaboration might be the most

important and domain-independent feature of tangible interaction, but this issue has attracted little explicit attention." Despite some good work in this area over the last couple of decades, there is still an opportunity for the research community to publish more integrated work that focuses on the 'social aspects' of tangible interaction, and how tangible design can facilitate social interaction for older adults.

5 DISCUSSION

This paper is a preliminary review to explore the design space of tangible user interfaces for older adults. Our research questions are driven by the need to understand the tangible design space for older adults' social interaction, and how they offer distinct advantages over existing digital technologies/screen-based interfaces. We found only 23 relevant papers where tangible technologies were used to enhance social connection for older adults. Although Ishii and Ullmer [77] introduced the idea of tangible user interfaces 20 years back, the results indicate the lack of evidence in research in using tangible technologies to improve social connection among older adults. Aging causes an inevitable degenerative effect on cognitive and motor capabilities of older adults [125], making it difficult for older adults to learn new technologies or use non-intuitive interfaces.

We present a summary of themes that emerged from our preliminary review to suggest design considerations for HCI researchers designing tangible technologies for older adults, which could result in lowering barriers and better technology adoption. While the first three design considerations are specific to tangible design for social connection, the rest address tangible design for older adults. We suggest that more integrated research from the HCI community addressing these design considerations of tangible user interfaces will not only address age-related bias in technology design and improve technology adoption among older adults, but also cater to the needs of older adults to stay socially connected.

Two-way exchange. Social Exchange Theory [72] states that, declining status, withdrawal, and isolation occurs from an unequal exchange of "investments and returns". When designing for older adults, HCI researchers should bank on the idea of 'social capital', which is embedded in one's social connections, where people share reciprocity, trust, and values. While family members want to take care of older members, older adults are also interested in learning about the well-being of their loved ones. Reciprocity or a two-way information-sharing is important to cater to older adults' need for checking in on their loved ones.

Asynchronous communication. Traditional wired telephone systems or landlines are still one of the most commonly used way for interpersonal communication among older adults [16]. However, this requires synchronization in timing and immediate responses from people on both sides of the channel, which is not always a practical option because of differences in schedule or time zones. Being unaware of availability, older adults want to avoid disturbing their loved ones and this affects interpersonal communication [103]. Enabling asynchronous exchange would avoid this issue and allow a convenient way of information sharing.

Active participation. Technology design for older adults often do not allow older adults to be the end users of the technology. Technologies for older adults are often built for their family members or caregivers to allow passive monitoring or check-in [74, 106]. One-way passive monitoring of older adults takes away agency, reduces self-esteem and independence, and gives rise to social stigma [97, 130]. By giving back control over how older adults use the system nurtures their sense of autonomy. Technologies for social communication should allow older adults to be active users instead of being passively monitored by family or caregivers.

Aesthetics. Age-related declines affect an older adult's interaction with their environment, but do not affect their aesthetic sensibilities. Tangible interfaces have the ability to transform everyday objects into functional technologies, and therefore new technologies for older adults can be designed keeping their aesthetic needs in mind. Aesthetically-pleasing systems easily blend into the surroundings. Older adults value technology that are personalized. To this end, customizability in tangible systems could be a valuable feature that can offer tailored solutions. Maker electronics [78] and e-Textiles [111] could offer promising solutions.

Familiarity and intuitiveness. Older adults prefer using systems that are simple and intuitive. Intuitiveness stems from a sense of familiarity. Older adults find screen-based interfaces non-intuitive and hard to understand. Embedding functionality into a well-known, familiar, everyday object as a metaphor [1, 110], can reduce fear of new technology, the added stress of understanding the working principles of the system, and facilitate a shorter learning curve. This can also effectively reduce time, money, and effort invested in training older adults about the usage of a new technology.

Habits and competences. We suggest HCI researchers to design tangible technologies leveraging existing habits and traditional practices of older adults [29]. This can ensure a more effective, easier, and quicker integration of a new technology into their lifestyles and daily routine. When a technology is non-intrusive and embedded in everyday actions such as, making tea in the morning [54, 135], or opening the door to pick up newspaper, their adoption and integration become easier, because they meet values and needs of older adults.

Accessibility. Researchers have attempted to improve accessibility of screen-based interfaces with big childlike buttons and fonts [113], and modern digital systems still do not reflect that, because they follow a "one for all" approach and are not designed keeping the needs of older adults in mind. However, tangible interfaces provide better accessibility since they are physical interfaces that leverage natural gestures and affordances for interaction.

6 LIMITATIONS

This paper is a first step towards opening a conversation about design considerations for tangible technologies for older adults. The preliminary review in this paper is restricted by our search terms and not exhaustive. Other variations of the search keywords might yield different results. This review is instead an initial work towards exploring the design space for tangible interfaces for older

adults. This is first step towards a more extensive literature review from other databases beyond ACM DL. Our search keywords are not exhaustive. Also, the literature mentioned in this report might not cover all concerns involving social isolation. The paper focuses more on social well-being and might not cover all aspects of medical well-being, which could also lead to isolation. Nevertheless, the paper presents current state-of-the-art and trends in this area of research, explores the potential for tangible interface design while designing for older adults. More integrated work in future on using tangible system to address social isolation among older adults could potentially result in promising solutions to some of the discussed issues in the field.

7 CONCLUSION

This late-breaking work is a preliminary review to investigate the benefits of tangible user interfaces for older adults in the context of improving social connection. To this end, we have reviewed existing literature, investigated the potential benefits of tangible technologies for older adults, and suggested design considerations for HCI researchers designing tangible technologies to foster social communication of older adults with others. We have indicated the need for more integrated research in this area to improve technology adoption and reduce isolation among older adults.

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